

Lecture 8

initial-value problems
single-step numerical time integration methods
stability of numerical time integration

theory of initial-value problems

Definition: initial-value problem

Find a function $y(t)$ which satisfies

$$\begin{aligned}\frac{dy}{dt} &= f(t, y), \quad a \leq t \leq b, \\ y(a) &= y_a,\end{aligned}$$

where $f(t, y)$ is a given function and y_a is a given constant – the initial value of $y(t)$.

The notation $f(t, y)$ means $f(t, y(t))$. For example, $f(t, y) = t^2 \sin(y(t))$.

Definition: well-posed initial-value problem

The initial value problem

$$\begin{aligned}\frac{dy}{dt} &= f(t, y), \quad a \leq t \leq b, \\ y(a) &= y_a,\end{aligned}$$

is well-posed, if its solution $y(t)$ exists, is unique and depends continuously on the data, i.e., on a , y_a and f .

Definition: Lipschitz continuity

A function $f(t, y)$ is Lipschitz continuous in y on a set $D \subset \mathbb{R}^2$ if there exists a constant $L > 0$ such that:

$$|f(t, y_1) - f(t, y_2)| \leq L|y_1 - y_2|,$$

for all $(t, y_1), (t, y_2) \in D$.

The constant L is called the Lipschitz constant.

Here t and y in $f(t, y)$ are considered as two independent variables. Lipschitz continuity is a stronger requirement than the usual continuity.

Theorem: Differentiability and Lipschitz continuity

If the function $f(t, y)$ is differentiable with respect to y and there exists $L > 0$ such that

$$\left| \frac{\partial f(t, y)}{\partial y} \right| \leq L, \quad \text{for all } (t, y) \in D,$$

then the function $f(t, y)$ is Lipschitz continuous in y .

Theorem: conditions for well-posedness of initial-value problem

Let $D = \{(t, y) \mid a \leq t \leq b, -\infty < y < \infty\}$ and let $f(t, y)$ be a continuous function. If $f(t, y)$ is also Lipschitz continuous in y on D , then the initial-value problem

$$\begin{aligned}\frac{dy}{dt} &= f(t, y), \quad a \leq t \leq b, \\ y(a) &= y_a,\end{aligned}$$

is well posed.

idea of the numerical time integration of ODE's

Example: $f(t, y) = f(t) = \sin(t)$

$$\begin{aligned}\frac{dy}{dt} &= \sin(t), \quad 0 \leq t \leq T, \\ y(0) &= 0.\end{aligned}$$

Integrating both sides over t :

$$\begin{aligned}\int_0^t \frac{dy}{dt'} dt' &= \int_0^t \sin(t') dt', \\ y(t) - y(0) &= [-\cos(t')]_0^t = -\cos(t) + \cos(0), \\ y(t) &= 1 - \cos(t).\end{aligned}$$

Example: $f(t, y) = -y^2 \sin(t)$

$$\begin{aligned}\frac{dy}{dt} &= -y^2 \sin(t), \quad 0 \leq t \leq T, \\ y(0) &= 1.\end{aligned}$$

Separating the t and y variables, and integrating both sides:

$$\begin{aligned}\frac{1}{y^2} dy &= -\sin(t) dt, \\ \int_{y(0)}^{y(t)} \frac{1}{y^2} dy &= -\int_0^t \sin(t') dt', \\ \left[-\frac{1}{y} \right]_{y(0)}^{y(t)} &= [\cos(t')]_0^t, \\ -\frac{1}{y(t)} + 1 &= \cos(t) - 1 \\ y(t) &= \frac{1}{2 - \cos(t)}\end{aligned}$$

When analytical methods fail?

$f(t, y) = e^{-t^2}$, integral over t is not known

$f(t, y) = t \cos(y^2)$, integral over y is not known

$f(t, y) = \sin(ty^2)$, non-separable equation

Numerical time integration

- Split the time interval $[t_0, T]$ in pieces: $[t_0, t_1], [t_1, t_2], \dots, [t_N, T]$
Suppose that for each $[t_n, t_{n+1}]$, $y(t_n)$ is given. To find $y(t_{n+1})$ do:

1

$$\int_{t_n}^{t_{n+1}} \frac{dy}{dt} dt = \int_{t_n}^{t_{n+1}} f(t, y) dt$$
$$y(t_{n+1}) - y(t_n) = \int_{t_n}^{t_{n+1}} f(t, y) dt$$

- 2 Approximate integral over $f(t, y)$ by a numerical quadrature rule.
- 3 Solve the resulting algebraic equation for $y(t_{n+1})$.

elementary single-step time integration methods

We use notation $y_n = y(t_n)$, and uniform time stepping $t_{n+1} - t_n = \Delta t$, $n = 0, \dots, N$. Single-step methods involve only y_n and y_{n+1} in each step.

Forward Euler Method

Single-point numerical approximation of the integral:

$$y_{n+1} - y_n = \int_{t_n}^{t_{n+1}} f(t, y) dt \approx \Delta t f(t_n, y_n)$$

Denoting by w_n the numerical approximation of y_n :

$$w_{n+1} = w_n + \Delta t f(t_n, w_n), \quad n = 0, 1, \dots, N$$

This is an *explicit* method, where w_{n+1} is directly obtained from w_n without the need to solve an additional equation.

Backward Euler Method

Single-point numerical approximation of the integral:

$$y_{n+1} - y_n = \int_{t_n}^{t_{n+1}} f(t, y) dt \approx \Delta t f(t_{n+1}, y_{n+1})$$

Denoting by w_n the numerical approximation of y_n :

$$w_{n+1} = w_n + \Delta t f(t_{n+1}, w_{n+1}), \quad n = 0, 1, \dots, N$$

This is an *implicit* method. The above (possibly nonlinear) equation must be solved for w_{n+1} at each time step.

Trapezoidal Method

Two-point numerical approximation of the integral:

$$y_{n+1} - y_n = \int_{t_n}^{t_{n+1}} f(t, y) dt \approx \frac{\Delta t}{2} f(t_n, y_n) + \frac{\Delta t}{2} f(t_{n+1}, y_{n+1})$$

Denoting by w_n the numerical approximation of y_n :

$$w_{n+1} = w_n + \frac{\Delta t}{2} f(t_n, w_n) + \frac{\Delta t}{2} f(t_{n+1}, w_{n+1}), \quad n = 0, 1, \dots, N$$

This is an *implicit* method. The above (possibly nonlinear) equation must be solved for w_{n+1} at each time step.

Modified Euler Method

Two-point numerical approximation of the integral:

$$y_{n+1} - y_n = \int_{t_n}^{t_{n+1}} f(t, y) dt \approx \frac{\Delta t}{2} f(t_n, y_n) + \frac{\Delta t}{2} f(t_{n+1}, y_{n+1})$$

Denoting by w_n the numerical approximation of y_n :

$$\bar{w}_{n+1} = w_n + \Delta t f(t_n, w_n), \quad (\text{predictor step})$$

$$w_{n+1} = w_n + \frac{\Delta t}{2} f(t_n, w_n) + \frac{\Delta t}{2} f(t_{n+1}, \bar{w}_{n+1}), \quad (\text{corrector step})$$

First, forward Euler method is used to predict $\bar{w}_{n+1}$. Then, a corrected value w_{n+1} is obtained with the Trapezoidal formula. This is an *explicit* method, since $f(t_{n+1}, \bar{w}_{n+1})$ is known, if $\bar{w}_{n+1}$ is known.

stability of initial-value problems

Consider the problem

$$\frac{dy}{dt} = f(t, y), \quad t > t_0; \quad y(0) = y_0$$

and its perturbation:

$$\frac{d\tilde{y}}{dt} = f(t, \tilde{y}), \quad t > t_0; \quad \tilde{y}(0) = y_0 + \epsilon_0,$$

where ϵ_0 is the error in the initial data. The difference between the solutions is defined as:

$$\epsilon(t) = \tilde{y}(t) - y(t)$$

This is the error due to the perturbation of the initial data.

Stability of initial-value problem

The initial value problem is called *stable* if

$$|\epsilon(t)| < \infty, \quad \text{for all } t > t_0,$$

and *absolutely stable* if

$$\lim_{t \rightarrow \infty} |\epsilon(t)| = 0.$$

If $|\epsilon(t)|$ is unbounded the problem is *unstable*.

Stability of the *test equation*

Let $f(t, y) = \lambda y(t) + g(t)$. Then, the so-called *test equation* is:

$$\frac{dy}{dt} = \lambda y + g, \quad t > t_0; \quad y(0) = y_0,$$

and its perturbation:

$$\frac{d\tilde{y}}{dt} = \lambda \tilde{y} + g, \quad t > t_0; \quad \tilde{y}(0) = y_0 + \epsilon_0.$$

The difference $\epsilon(t) = \tilde{y}(t) - y(t)$ satisfies its own initial-value problem (why?):

$$\frac{d\epsilon}{dt} = \lambda \epsilon, \quad t > t_0; \quad \epsilon(0) = \epsilon_0,$$

which can be solved as (how?):

$$\epsilon(t) = \epsilon_0 e^{\lambda(t-t_0)}.$$

Stability of the *test equation*: real and complex cases

Perturbation $\epsilon(t)$ of the solution of the test problem evolves as:

$$\epsilon(t) = \epsilon_0 e^{\lambda(t-t_0)}.$$

- Let $\lambda \in \mathbb{R}$. Then, the test problem is stable if $\lambda \leq 0$, absolutely stable if $\lambda < 0$, unstable if $\lambda > 0$.
- Let $\lambda \in \mathbb{C}$, i.e., $\lambda = \operatorname{Re}(\lambda) + i\operatorname{Im}(\lambda)$. Then,

$$\begin{aligned} |\epsilon(t)| &= \left| \epsilon_0 e^{(\operatorname{Re}(\lambda) + i\operatorname{Im}(\lambda))(t-t_0)} \right| = |\epsilon_0| \left| e^{\operatorname{Re}(\lambda)(t-t_0)} \right| \left| e^{i\operatorname{Im}(\lambda)(t-t_0)} \right| \\ &= |\epsilon_0| e^{\operatorname{Re}(\lambda)(t-t_0)}, \end{aligned}$$

and the test problem is stable if $\operatorname{Re}(\lambda) \leq 0$, absolutely stable if $\operatorname{Re}(\lambda) < 0$, unstable if $\operatorname{Re}(\lambda) > 0$.

stability of numerical time integration methods

Numerical solution of the test equation: FE Method

Forward Euler Method for the test equation:

$$w_{n+1} = w_n + \Delta t (\lambda w_n + g_n) = (1 + \lambda \Delta t) w_n + \Delta t g_n, \quad w_0 = y_0.$$

Forward Euler Method for perturbed test equation:

$$\tilde{w}_{n+1} = \tilde{w}_n + \Delta t (\lambda \tilde{w}_n + g_n) = (1 + \lambda \Delta t) \tilde{w}_n + \Delta t g_n, \quad \tilde{w}_0 = y_0 + \epsilon_0$$

Numerical approximation of $\epsilon(t_n)$, i.e., $\bar{\epsilon}_n = \tilde{w}_n - w_n$, satisfies:

$$\bar{\epsilon}_{n+1} = (1 + \lambda \Delta t) \bar{\epsilon}_n, \quad \bar{\epsilon}_0 = \epsilon_0.$$

Numerical solution of the test equation: BE Method

Backward Euler Method for the test equation:

$$w_{n+1} = w_n + \Delta t (\lambda w_{n+1} + g_{n+1}), \quad w_0 = y_0,$$
$$w_{n+1} = \frac{1}{1 - \lambda \Delta t} w_n + \frac{\Delta t g_{n+1}}{1 - \lambda \Delta t}, \quad w_0 = y_0,$$

Backward Euler Method for perturbed test equation:

$$\tilde{w}_{n+1} = \frac{1}{1 - \lambda \Delta t} \tilde{w}_n + \frac{\Delta t g_{n+1}}{1 - \lambda \Delta t}, \quad \tilde{w}_0 = y_0 + \epsilon_0,$$

Numerical approximation of $\epsilon(t_n)$, i.e., $\bar{\epsilon}_n = \tilde{w}_n - w_n$, satisfies:

$$\bar{\epsilon}_{n+1} = \frac{1}{1 - \lambda \Delta t} \bar{\epsilon}_n, \quad \bar{\epsilon}_0 = \epsilon_0.$$

Amplification factors of single-step methods

For all single-step methods the numerical approximation of the perturbation has the form:

$$\bar{\epsilon}_{n+1} = Q \bar{\epsilon}_n$$

with the following *amplification factors*:

$$Q = 1 + \lambda \Delta t, \quad (\text{Forward Euler})$$

$$Q = \frac{1}{1 - \lambda \Delta t}, \quad (\text{Backward Euler})$$

$$Q = \frac{1 + \frac{1}{2}\lambda \Delta t}{1 - \frac{1}{2}\lambda \Delta t}, \quad (\text{Trapezoidal})$$

$$Q = 1 + \lambda \Delta t + \frac{1}{2}(\lambda \Delta t)^2, \quad (\text{Modified Euler})$$

Stability of numerical methods for test equation

Since

$$\bar{\epsilon}_{n+1} = Q \bar{\epsilon}_n = Q^n \epsilon_0,$$

and

$$|\bar{\epsilon}_{n+1}| = |Q^n \epsilon_0| = |Q|^n |\epsilon_0|,$$

a numerical method is *stable*, i.e. $|\bar{\epsilon}_n| < \infty$ for all $n = 0, 1, \dots$, if

$$|Q| \leq 1,$$

absolutely stable, i.e. $\lim_{n \rightarrow \infty} |\bar{\epsilon}_n| = 0$, if

$$|Q| < 1,$$

and *unstable*, i.e. $|\bar{\epsilon}_n|$ is unbounded, if

$$|Q| > 1.$$

Stability condition for the Forward Euler Method

Assume that the initial-value problem is absolutely stable, i.e. $\lambda < 0$ or $\lambda = -|\lambda|$ and $|\lambda| > 0$. Then, the Forward Euler Method, $Q = 1 + \lambda\Delta t$, is stable if:

$$\begin{aligned} |Q| &\leq 1, \\ -1 &\leq Q \leq 1, \\ -1 &\leq 1 + \lambda\Delta t \leq 1, \\ -1 &\leq 1 - |\lambda|\Delta t \leq 1, \\ -2 &\leq -|\lambda|\Delta t \leq 0, \\ 0 &\leq \Delta t \leq \frac{2}{|\lambda|}. \end{aligned}$$

Stability condition: $\Delta t \leq 2/|\lambda|$.

Forward Euler Method is only *conditionally stable*.

Stability of the Backward Euler Method

The Backward Euler Method's amplification factor is $Q = 1/(1 - \lambda\Delta t)$.

With $\lambda = -|\lambda|$, $|\lambda| > 0$, we have:

$$\begin{aligned} |Q| &\leq 1, \\ -1 &\leq Q \leq 1, \\ -1 &\leq \frac{1}{1 - \lambda\Delta t} \leq 1, \\ -1 &\leq \frac{1}{1 + |\lambda|\Delta t} \leq 1 \end{aligned}$$

This holds for any $\Delta t > 0$.

Backward Euler Method is *unconditionally stable*.

Stability of general problems

For a general $f(t, y)$

$$\frac{dy}{dt} = f(t, y), \quad t > t_0; \quad y(0) = y_0,$$

consider the Taylor series around $\hat{t}$ and $\hat{y}$:

$$\begin{aligned} f(t, y) &= f(\hat{t}, \hat{y}) + \frac{\partial f(\hat{t}, \hat{y})}{\partial y}(y - \hat{y}) + \frac{\partial f(\hat{t}, \hat{y})}{\partial t}(t - \hat{t}) + C_1(t - \hat{t})^2 + C_2(y - \hat{y})^2 \\ &= \lambda y + g(t) + C_2(y - \hat{y})^2. \end{aligned}$$

Neglecting $C_2(y - \hat{y})^2$, this can be reduced to the test equation with $f(t, y) = \lambda y + g(t)$ and

$$\lambda = \frac{\partial f}{\partial y}(\hat{t}, \hat{y}).$$

Approximate (local) stability condition $\lambda < 0$, i.e., $\frac{\partial f}{\partial y}(\hat{t}, \hat{y}) < 0$, depends on $(\hat{t}, \hat{y})$.

Example: analytical stability

Consider the initial-value problem

$$\frac{dy}{dt} = -10y^2 + 20, \quad t > 0; \quad y(0) = 0.$$

This is a separable equation with the analytical solution:

$$y(t) = \frac{e^{t20\sqrt{2}} - 1}{e^{t20\sqrt{2}} + 1} \sqrt{2}.$$

Linearization (first two terms of Taylor series) of $f(t, y)$ around $(\hat{t}, \hat{y})$:

$$\begin{aligned} -10y^2 + 20 &= -10\hat{y}^2 + 20 + \left. \frac{\partial}{\partial y} (-10y^2 + 20) \right|_{y=\hat{y}} (y - \hat{y}) + \mathcal{O}((y - \hat{y})^2) \\ &= -10\hat{y}^2 + 20 - 20\hat{y}(y - \hat{y}) + \mathcal{O}((y - \hat{y})^2) \\ &= \lambda y + g(t) + \mathcal{O}((y - \hat{y})^2), \end{aligned}$$

where

$$\lambda = -20\hat{y}.$$

The problem is locally stable ($\lambda < 0$) for $\hat{y} > 0$.

Example: numerical stability of the FE method

Applying the Forward Euler method with w_i denoting the numerical approximation of y_i :

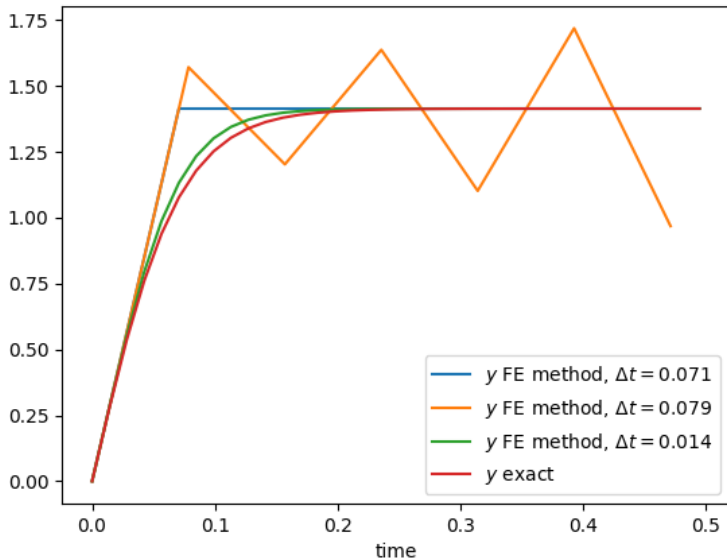
$$\begin{aligned}w_0 &= 0, \\w_{i+1} &= w_i + \Delta t (-10w_i^2 + 20), \quad i = 0, 1, \dots\end{aligned}$$

Stability condition on Δt for the FE method:

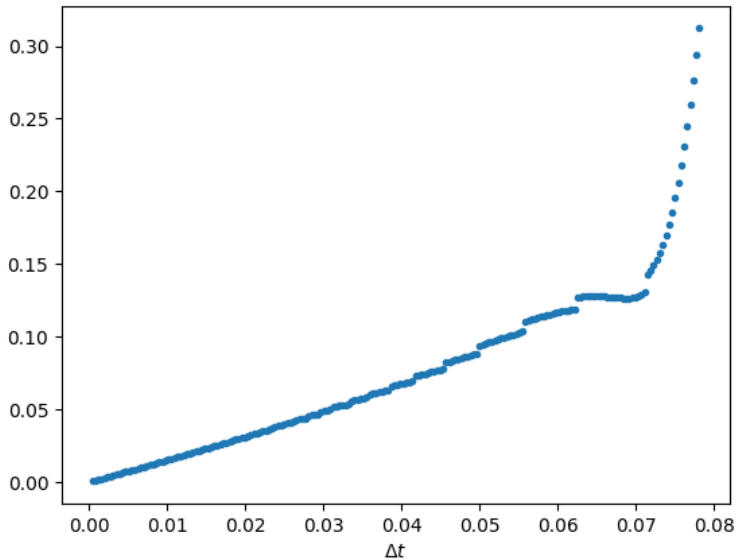
$$\Delta t \leq \frac{2}{|\lambda|} = \frac{2}{20|\hat{y}|} \leq \frac{2}{20 \max_t |y(t)|} = \frac{1}{10\sqrt{2} \max_t |\tanh(t\sqrt{300})|} = \frac{1}{10\sqrt{2}}$$

In general, $y(t)$ is not known, and a stable Δt is obtained experimentally (educated trial and error). Instability is easy to spot.

stability condition: $\Delta t \leq 0.071$



global error (RMSE)



Questions

- Derive the amplification factors for the Trapezoidal and Modified Euler Methods applied to the test equation.
- Study the stability of the Trapezoidal and Modified Euler Methods for the test equation, either deriving the stability condition on the time step Δt or showing that the method is unconditionally stable.